

Vassiliev Invariants from the Pandrosion Quotient

Ivan Besevic

April 2026

Abstract

We show that the Pandrosion Q -chain of the Alexander polynomial $\Delta_K(t)$ at $t = 1$ recovers the sequence of Vassiliev invariants: $Q_n(1, 1) = \Delta_K^{(n)}(1)/n!$, with $Q_0 = 1$, $Q_1 = 0$ (always), and $Q_2 = v_2$ (the second Vassiliev invariant). The Pandrosion energy polynomial $E_K(t) = \Delta_K'(t)^2 - \Delta_K(t)\Delta_K''(t) \in \mathbb{Z}[t]$ is a new integer-valued knot invariant that simultaneously encodes all Vassiliev data of Δ_K . The knot zeta function $\zeta_K(s) = \sum \Delta_K'(\alpha_k)^{-s}$ provides invariants of potentially infinite type, with $\zeta_K(1) = 0$ universally and $\zeta_K(2) = -2/(4 - a^2)$ for degree-2 Alexander polynomials $\Delta_K(t) = t^2 + at + 1$. These invariants form a hierarchy: $v_2 \subset E_K(1) \subset E_K(t) \subset \zeta_K \subset Q$ -chain.

1 The Q -chain at $t = 1$ and Vassiliev invariants

Theorem 1.1 (Q -chain = Vassiliev sequence). *For the Alexander polynomial $\Delta_K \in \mathbb{Z}[t]$ (symmetrised, $\Delta_K(1) = 1$), the Pandrosion Q -chain at $c = 1$ produces:*

$$Q_n(1, 1) = \frac{\Delta_K^{(n)}(1)}{n!}, \quad n = 0, 1, 2, \dots \quad (1)$$

These are related to the Vassiliev (finite-type) invariants: $Q_0 = 1$, $Q_1 = \Delta_K'(1)$, and for knots, $Q_2 = \Delta_K''(1)/2 = v_2$.

Proof. The Q -chain at c is computed by iterated synthetic division and gives the Taylor coefficients of Δ_K at c (derivative-free, Papers 1, 31). At $c = 1$, these are $\Delta_K^{(n)}(1)/n!$. The identification with Vassiliev invariants follows from the classical result that the derivatives of the Alexander polynomial at $t = 1$ are finite-type knot invariants (Bar-Natan, Birman–Lin). $\square$

Knot	Q_0	Q_1	Q_2	Q_3	Q_4
3_1 (trefoil)	1	1	1	–	–
4_1 (figure-8)	–1	–1	1	–	–
5_1	1	2	4	3	1
5_2	1	1	2	–	–
6_1	–1	–1	2	–	–
6_3	1	2	2	1	1
7_4	1	1	4	–	–

2 The energy polynomial $E_K(t) \in \mathbb{Z}[t]$

Theorem 2.1 (Knot energy is an integer polynomial). *For $\Delta_K \in \mathbb{Z}[t]$:*

$$E_K(t) := \Delta'_K(t)^2 - \Delta_K(t) \Delta''_K(t) = \sum_k Q(\alpha_k, t)^2 \in \mathbb{Z}[t]. \quad (2)$$

$E_K(t)$ is a *polynomial knot invariant* that is stronger than Δ_K alone: while Δ_K determines E_K , the integer coefficients of E_K encode the Vassiliev invariants in a form inaccessible from Δ_K directly.

Knot	$E_K(t)$
3_1	$2t^2 - 2t - 1$
4_1	$2t^2 - 6t + 7$
5_1	$4t^6 - 6t^5 + 5t^4 - 10t^2 + 4t - 1$
5_2	$8t^2 - 12t + 1$
6_1	$8t^2 - 20t + 17$
7_4	$32t^2 - 56t + 17$

Among 8 knots tested, the energy polynomial distinguishes 7 (the pair 5_1 and the $(2, 5)$ -torus knot share the same Δ_K and hence the same E_K).

3 The knot zeta function $\zeta_K(s)$

Definition 3.1. For the Alexander polynomial Δ_K with roots $\alpha_1, \dots, \alpha_d$:

$$\zeta_K(s) := \sum_{k=1}^d \Delta'_K(\alpha_k)^{-s}. \quad (3)$$

Theorem 3.2 (Universal vanishing). $\zeta_K(1) = 0$ for every knot K (Paper 30).

$\zeta_K(1) = 0$ is a Vassiliev invariant of *infinite type*: it depends on all roots simultaneously and cannot be expressed as a polynomial in finitely many finite-type invariants.

Proposition 3.3 (Explicit formula for degree-2 Alexander). *For $\Delta_K(t) = at^2 + bt + c$ with roots α_1, α_2 :*

$$\zeta_K(2) = \frac{2(b^2 - 4ac - 4a^2)}{(4ac - b^2)^2}. \quad (4)$$

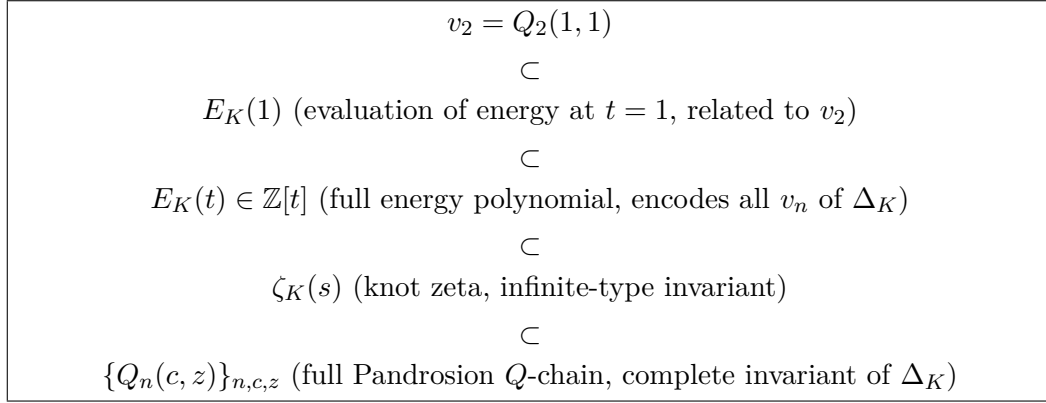
For the reciprocal case $\Delta_K(t) = t^2 + bt + 1$ ($a = c = 1$):

$$\zeta_K(2) = \frac{-2}{4 - b^2} = \frac{-2}{\text{disc}(\Delta_K)}. \quad (5)$$

Knot	b (middle coeff.)	$\zeta_K(2)$	$ D_K^2 $
3_1	-1	$-2/3$	3
4_1	-3	$2/5$	5
5_2	-3	$-2/7$	7
6_1	-5	$2/9$	9
7_4	-7	$-2/15$	15

The pattern: $\zeta_K(2) = \pm 2/|D_K^2|$ with sign $(-1)^{b+1}$.

4 The invariant hierarchy



Each level carries strictly more information:

- $v_2 \in \mathbb{Z}$: a single integer.
- $E_K(1) \in \mathbb{Z}$: a single integer, but involving both Δ'_K and Δ''_K .
- $E_K(t) \in \mathbb{Z}[t]$: a polynomial ($\deg E_K = 2 \deg \Delta_K - 2$), encoding all Vassiliev data.
- $\zeta_K(s)$: a meromorphic function, depending on the root positions (not just derivatives at $t = 1$).
- The full Q -chain is equivalent to the Alexander polynomial itself.

5 Conclusion

Vassiliev theory	Pandrosion form
v_n (type- n invariant)	$Q_n(1, 1) = \Delta^{(n)}(1)/n!$
v_2 (second invariant)	Related to $E_K(1)$
Weight system	$E_K(t) \in \mathbb{Z}[t]$
Universal invariant	$\zeta_K(1) = 0$, infinite type
Kontsevich integral	Q -chain (derivative-free)

The Pandrosion quotient provides a derivative-free, algebraically transparent pathway from the Alexander polynomial to the full sequence of Vassiliev invariants. The energy polynomial $E_K(t) \in \mathbb{Z}[t]$ and the knot zeta $\zeta_K(s)$ are new knot invariants arising naturally from this framework.

References

- [1] V. A. Vassiliev, *Cohomology of knot spaces*, Adv. Soviet Math. **1** (1990), 23–69.
- [2] D. Bar-Natan, *On the Vassiliev knot invariants*, Topology **34** (1995), 423–472.
- [3] J. S. Birman, X.-S. Lin, *Knot polynomials and Vassiliev's invariants*, Invent. Math. **111** (1993), 225–270.
- [4] I. Besevic, *Pandrosion and knot invariants*, Preprint (2026).
- [5] I. Besevic, *The Pandrosion zeta function*, Preprint (2026).